JOHANNES J. Z. STRAUSS | 党壮壮

Ph.D. Student in Mathematics

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RESEARCH INTERESTS

I am a Ph.D. student in Mathematics at the Shanghai Institute for Mathematics and Interdisciplinary Sciences (SIMIS), Fudan University. My research spans computational medicine, biomathematics, biostatistics, and public health. Broadly, I develop statistical, mathematical, and computational methods for analyzing high-dimensional data, with an emphasis on omics data (transcriptomics and multi-omics).

Keywords: Bayesian Inference · Statistical Genetics · Omics Data · Machine & Statistical Learning · High-Dimensional Data · Bioinformatics

EDUCATION

Ph.D., Mathematics 2026 – present
Shanghai Institute for Mathematics & Interdisciplinary Sciences (SIMIS) | *Fudan University* Shanghai, P. R. China

- Supervisors:
  - Prof. Bingjie Li (李冰杰), Tongji Medical College, Huazhong University of Science & Technology
  - Prof. Zhigang Yao (姚志刚), SIMIS & National University of Singapore

M.S., Statistics 2025
University of Colorado Denver | *Graduate School* Denver, CO, USA

- Advisors: Yaning Liu, Ph.D. & Anna Peljto, Dr.P.H.
- Project: *"Enhancing Genetic Risk Prediction: Leveraging Sparse Variational Bayesian Inference to Model Polygenic Risk in Idiopathic Pulmonary Fibrosis"*

B.S., Mathematics 2024
University of Colorado Denver | *College of Liberal Arts & Sciences* Denver, CO, USA

RESEARCH FELLOWSHIPS & PROFESSIONAL EXPERIENCE

Graduate Research Assistant Fall 2026 – present
Applied Statistics & Computational Medicine Team | *SIMIS, Fudan University* Shanghai, P. R. China

Graduate Research Assistant Fall 2026 – present
Laboratory of Bingjie Li, Ph.D. | *School of Public Health, Tongji Medical College, Huazhong University of Science & Technology* Wuhan, Hubei, P. R. China

Bioinformatician (Temporary) Jan. – Aug. 2026
Laboratory of Martin Breuss, Ph.D. | *University of Colorado School of Medicine* Aurora, CO, USA

- Department of Pediatrics, Section of Genetics & Metabolism

PATH-GDS Graduate Research Fellow 2024 – 2025
Pathways in Genomic Data Science | *University of Colorado Anschutz Medical Campus* Aurora, CO, USA

- NIH Grant #R25HG012919 · hendricks-research-team.github.io/PATHGDS
- Laboratory rotations:

- Laboratory of Martin Breuss, Ph.D. — School of Medicine, Department of Pediatrics, Section of Genetics & Metabolism
- Laboratory of David Schwartz, M.D. & Ivana Yang, Ph.D. — School of Medicine, Department of Medicine, Division of Pulmonary, Allergy & Critical Care Medicine

Abstracts, Conferences & Presentations

NHGRI Research Training & Career Development Annual Meeting Apr. 7–8, 2025
Poster presentation Philadelphia, PA, USA

- *"Enhancing Genetic Risk Prediction: Leveraging Sparse Variational Bayesian Inference to Model Polygenic Risk in Idiopathic Pulmonary Fibrosis"*

American Society of Human Genetics (ASHG) Annual Meeting Nov. 5–9, 2024
Presentation Denver, CO, USA

Grants, Scholarships & Funding

Chinese Government Scholarship 2026
China Scholarship Council, Ministry of Education, Beijing Fully funded Ph.D.

PATH-GDS Fellowship 2024 – 2025
Dept. of Biomedical Informatics, University of Colorado School of Medicine Fully funded M.S. & NIH fellowship

Departmental Scholarship 2024
Dept. of Mathematical & Statistical Sciences, University of Colorado Denver $1,500

Travel Awards 2025
Dept. of Mathematical & Statistical Sciences; College of Liberal Arts & Sciences; Office of Graduate Education, University of Colorado Denver $1,500 total

Journal Review Service

Ad Hoc Reviewer: *Pediatric Obesity* (2026) · *Molecular Therapy – Nucleic Acids* (2026)

Technical Skills

Languages	R, Python, MATLAB
Computing	Linux/Unix-based HPC clusters, command-line tools
Bioinformatics	Standard pipelines and toolkits (GATK, bedtools, etc.)

Professional Memberships

American Society of Human Genetics (ASHG)

American Statistical Association (ASA)

International Chinese Statistical Association (ICSA)

Society for Industrial & Applied Mathematics (SIAM)